

VirtuBox Assessment: Software Tester (QA)

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1. Test Scenarios and Test Cases

I would test the application step by step. First I would check registration and login, and after that I would check the task related features. I would also check what happens when the user enters wrong, missing or unusual data.

Registration

Positive Cases

- I'd check that a user can register successfully when username, email and password are entered with valid details.
- Check that a valid email format is accepted (for example, **abc@gmail.com**, not **abc gmail.com**).
- Check that the registration completes successfully and the user can use the account for login.

Negative Cases

- I'd check that an invalid email format is rejected (for example, **abc gmail.com**).
- Check that registration is not allowed when a required field such as username, email or password is empty.
- Check that the same email cannot be used to create another account, if email uniqueness is required.
- Check that invalid or weak password input is rejected according to the application's password rules, if such rules are defined.

Edge Cases

- Check very long username, email or password input and make sure the application handles it properly.
- Check unusual but valid email characters and make sure valid input is not rejected unnecessarily.
- Check what happens when registration is submitted multiple times quickly.

Login

Positive Cases

- I'd check that a registered user can log in using the correct email and password.
- Check that the user is taken to the task section after successful login.

Negative Cases

- I'd Check that login fails when the password is incorrect.
- Check that login fails when the email is not registered.
- Check that an invalid email format is rejected (for example, **abc gmail.com** instead of **abc@gmail.com**).
- Check that login is not allowed when the email or password field is empty.

- Check that the Forgot Password option works properly, **if provided**.

Edge Cases

- Check very long or unusual input in the email and password fields.
- Check repeated wrong-password attempts and make sure the application handles them safely.

Create Task

Positive Cases

- Check that a task is created successfully when all required details are entered correctly.
- Check that a valid due date is accepted and the created task appears in the task list.
- Check that the task is saved correctly in the database.

Negative Cases

- Check that a task cannot be created when required task details are missing.
- Check that invalid date input is rejected.
- Assuming past due dates are not allowed, check that a past date is rejected.
- Check that the application shows an error if the database save fails instead of showing a false success message.

Edge Cases

- Check very long task details and make sure the application handles them properly.
- Check whether multiple quick clicks on Create produce duplicate tasks.

View Task List

Positive Cases

- Check that all created tasks are displayed in the task list.
- Check that the correct task details are shown.
- Check that a newly created task appears in the list.

Negative Cases

- Check that the application does not show incorrect or missing task information.
- Check that a deleted task no longer appears in the task list.

Edge Cases

- Check the task list when there are no tasks and make sure the page does not break.
- Check the task list when there are many tasks and make sure the list still loads properly.

Edit Task

Positive Cases

- Check that an existing task can be edited successfully.
- Check that the updated task details are shown correctly after saving.
- Check that the updated values are also saved in the database.

Negative Cases

- Check that invalid or missing required information is not saved while editing.
- Check that an invalid date, such as month 13 or 14, is rejected.

- Check that an error is shown if the updated task cannot be saved to the database.

Edge Cases

- Check saving without making any changes and make sure the task remains unchanged.
- Check editing a task with very long details.

Delete Task

Positive Cases

- Check that a task is deleted successfully after the user confirms deletion.
- Check that the deleted task no longer appears in the task list.
- Check that the database reflects the expected deletion.

Negative Cases

- Check that cancelling the delete confirmation keeps the task unchanged.
- Check that an error is shown if deletion fails instead of showing a false success message.

Edge Cases

- Check whether a confirmation message such as “Do you really want to delete this?” is shown before permanent deletion.
- Check multiple quick clicks on Delete and make sure the task is not processed incorrectly more than once.

2. Input Validation & Error Handling

I would make sure that wrong or incomplete input does not break the application. The user should get a simple message explaining what went wrong. For example, an invalid email should show an email related error and an empty required field should be pointed out.

Positive Cases

- Valid information should be accepted and the requested operation should complete successfully.
- After Create, Edit or Delete, the information shown in the application should match the expected database state.

Negative Cases

- Empty required fields should show a clear validation message instead of allowing invalid data to be saved.
- Invalid email, password, date or other invalid input should be rejected with a simple and understandable error message.
- If there is a technical/database problem, the application should show a message such as “We are currently having a technical glitch, please try again later.”
- The application should not show a success message when the requested operation actually fails.

Edge Cases

- Very long input should be handled properly without breaking the page or application.
- If one feature has a technical problem, the rest of the application should continue working where possible instead of the whole application going down.
- User data and created tasks should have proper backup/recovery support so that important data is not easily lost.

3. Potential Bugs / Risk Areas

Since the application has not been executed, these are possible bugs or risk areas that I would look for. I would not call them confirmed bugs until they are actually tested.

1. Same email can be used for more than one account

Severity: Major

Reason / Impact: A duplicate account can create confusion and may cause problems while logging in or identifying the correct user.

2. Valid registration fails

Severity: Major

Reason / Impact: A user may provide correct details but still be unable to create an account, which prevents access to the application.

3. Newly created task is not saved in the database

Severity: Critical

Reason / Impact: The user may see a success message, but the task can be lost. This affects the main purpose of the application.

4. Edit changes are not saved or old values are shown

Severity: Major

Reason / Impact: The user may think the task was updated, while the old information remains in the application or database.

5. New task does not appear in the task list

Severity: Major

Reason / Impact: The task may have been created but the user cannot see it, making the task-management feature unreliable.

6. Deleted task still appears in the task list

Severity: Major

Reason / Impact: The UI may show information that should no longer be available, creating confusion about whether deletion actually happened.

7. UI and database show different task information

Severity: Major

Reason / Impact: Different data in the application and database can lead to incorrect information and loss of trust in the system.

8. Delete happens without proper confirmation

Severity: Major

Reason / Impact: A user could accidentally delete a task, especially if the deletion is permanent and there is no way to undo it.

9. Existing account is later treated as unregistered

Severity: Critical

Reason / Impact: A user could lose access to their account and previous tasks. This is a serious issue because it affects both login and stored user data.

10. Wrong login credentials are accepted

Severity: Critical

Reason / Impact: An incorrect password or other invalid credentials being accepted can allow unauthorized access to another user's account and tasks.

4. Overall Approach

My approach would be to first check the main features with correct information and make sure the normal flow is working. After that I would try wrong, missing and unusual inputs. I would also check whether the information remains correct when moving from one feature to another, for example Create to View, Edit to View and Delete to View. I would give more attention to login, account access, saving data and data consistency because these can affect the user's work.

Note: Some checks such as CAPTCHA, username uniqueness, password strength, past-date restrictions and permanent deletion are not specifically defined in the assessment. I would confirm the actual application requirements before treating these as definite failures